Homework 5
Macroeconomic Theory I

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A permanent link to this assignment, plus a code repository, can be found at `https://github.com/AFB-TUV/Fall-2026/tree/main/Macro/Homework%205`.

1 CES: Profit Maximization

1.1

We want to solve the consumer's problem:

$$\max_{q(\omega)} Q = \left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}} \text{ s.t.} \int_\Omega p(\omega)q(\omega)d\omega \leqslant E$$

This lets us set up the Lagrangian:

$$\mathcal{L} = \left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}} + \lambda\left[E - \int_\Omega p(\omega)q(\omega)d\omega\right]$$

Now we take the first-order condition with respect to $q(\omega)$ and set it equal to zero. This gives

$$\cancel{\left(\frac{\sigma}{\sigma-1}\right)}\left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{1}{\sigma-1}} \cancel{\left(\frac{\sigma-1}{\sigma}\right)}\left(q(\omega)^{-\frac{1}{\sigma}}\right) - \lambda p(\omega) = 0$$

We can look at this first-order condition and recognize that the first surviving term is just

$Q^{\frac{1}{\sigma}}$. Substituting this and rearranging to solve for λ, we get

$$\lambda = \frac{1}{p(\omega)} q(\omega)^{-\frac{1}{\sigma}} Q^{\frac{1}{\sigma}}$$

Now we consider a second good ω', which should be subject to the same logic mutatis mutandis. Both first-order conditions are rearranged to be set equal to λ, so we can write

$$\frac{1}{p(\omega)} q(\omega)^{-\frac{1}{\sigma}} \cancel{Q^{\frac{1}{\sigma}}} = \frac{1}{p(\omega')} q(\omega')^{-\frac{1}{\sigma}} \cancel{Q^{\frac{1}{\sigma}}}$$

From here it is simple to rearrange this equation to give

$$\frac{q(\omega)}{q(\omega')} = \left(\frac{p(\omega)}{p(\omega')} \right)^{-\sigma}$$

as required.

1.2

Now we want to derive the aggregate price index. We eventually do this by integrating the price function for ω over the goods index Ω, but we first need to do it in such a way that we get rid of $q(\omega)$. We do this by integrating out ω such that it becomes the aggregate Q.

We start from our first-order condition rearranged for λ, and rearrange it to solve for $q(\omega)$:

$$\lambda = \frac{1}{p(\omega)} q(\omega)^{-\frac{1}{\sigma}} Q^{\frac{1}{\sigma}} \implies \lambda p(\omega) Q^{-\frac{1}{\sigma}} = q(\omega)^{-\frac{1}{\sigma}} \implies q(\omega) = \lambda^{-\sigma} p(\omega)^{-\sigma} Q$$

We now raise both sides to the power of $\frac{\sigma-1}{\sigma}$ to get

$$q(\omega)^{\frac{\sigma-1}{\sigma}} = \left[\lambda^{-\sigma} p(\omega)^{-\sigma} Q \right]^{\frac{\sigma-1}{\sigma}} = \lambda^{1-\sigma} p(\omega)^{1-\sigma} Q^{\frac{\sigma-1}{\sigma}}$$

Then we integrate:

$$Q^{\frac{\sigma-1}{\sigma}} = \int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega = \int_\Omega \lambda^{1-\sigma} p(\omega)^{1-\sigma} Q^{\frac{\sigma-1}{\sigma}} d\omega$$

We can pull the Q and λ terms out of the integral because they are independent of the quantity index ω, which gives us

$$\cancel{Q^{\frac{\sigma-1}{\sigma}}} = \cancel{Q^{\frac{\sigma-1}{\sigma}}}\lambda^{1-\sigma}\int_\Omega p(\omega)^{1-\sigma}d\omega \implies 1 = \lambda^{1-\sigma}\int_\Omega p(\omega)^{1-\sigma}d\omega$$

We now solve this for λ:

$$\lambda^{-(\sigma-1)} = \int_\Omega p(\omega)^{1-\sigma}d\omega \implies \lambda = \left[\int_\Omega p(\omega)^{1-\sigma}d\omega\right]^{-\frac{1}{1-\sigma}}$$

We can look at the resulting expression and recognize that it can be rewritten

$$\lambda = \left(\left[\int_\Omega p(\omega)^{1-\sigma}d\omega\right]^{\frac{1}{1-\sigma}}\right)^{-1} = \frac{1}{\left[\int_\Omega p(\omega)^{1-\sigma}d\omega\right]^{\frac{1}{1-\sigma}}}$$

The term in the denominator is recognizably a CES quantity, so let's call this the price index. Define it as

$$P \equiv \left(\int_\Omega p(\omega)^{1-\sigma}d\omega\right)^{\frac{1}{1-\sigma}}$$

so the equation becomes

$$\lambda = \frac{1}{P} = P^{-1}$$

Now, because we have integrated the original first-order condition, our solution for λ is now algebraically distinct from the original first-order condition. This allows us to substitute for λ in the (rearranged) first-order condition:

$$q(\omega) = \lambda^{-\sigma}p(\omega)^{-\sigma}Q = P^\sigma p(\omega)^{-\sigma}Q = \left(\frac{p(\omega)}{P}\right)^{-\sigma}Q$$

Now we derive a way to put expenditures E into this equation. Our equation for expenditures is the budget constraint $E = p(\omega)q(\omega)$, so we can get there by multiplying both sides by $p(\omega)$ and integrating:

$$E = \int_\Omega p(\omega)q(\omega)d\omega = \int_\Omega p(\omega)\left(\frac{p(\omega)}{P}\right)^{-\sigma} Q d\omega$$

Now we observe that the aggregate of prices, the bracketed expression P, will not depend on ω since we have already taken care of the ω by integrating over Ω! This lets us pull it (and Q) out of the integral:

$$E = \frac{Q}{P^{-\sigma}}\left(\int_\Omega p(\omega)p(\omega)^{-\sigma}d\omega\right) = \frac{Q}{P^{-\sigma}}\int_\Omega p(\omega)^{1-\sigma}d\omega$$

We recognize that the latter term is just $P^{1-\sigma}$, so we can finally write aggregate expenditures as

$$E = QP^{1-\sigma+\sigma} = PQ$$

which is what we want!

Now we observe that we obtained λ above as $\lambda = \frac{1}{P}$. This allows us to derive $q(\omega)$ as a function of P from the first-order condition:

$$q(\omega) = \lambda^{-\sigma}p(\omega)^{-\sigma}Q = \left(\frac{p(\omega)}{P}\right)^{-\sigma} Q$$

Multiplying the right-hand side by $1 = \frac{P}{P}$, we derive an equation in terms of total expenditures E:

$$q(\omega) = \left(\frac{p(\omega)}{P}\right)^{-\sigma}\frac{PQ}{P} = \left(\frac{p(\omega)}{P}\right)^{-\sigma}\frac{E}{P}$$

as required.

1.3

Now we consider the problem of the monopolistically competitive plan. It chooses a price $p(\omega)$ to solve

$$\max_{p(\omega)} \pi = \left(p(\omega) - \frac{w}{\phi} \right) q(\omega)$$

where ϕ represents the idiosyncratic productivity of the firm. We now have an equation for $q(\omega)$, so we can substitute it to get

$$\max_{p(\omega)} \left(p(\omega) - \frac{w}{\phi} \right) \left(\frac{p(\omega)}{P} \right)^{-\sigma} \frac{E}{P} = \frac{E}{P} \left[\frac{p(\omega)^{1-\sigma}}{P} - \frac{w p(\omega)^{-\sigma}}{\phi P} \right]$$

We derive the first-order condition as

$$0 = \frac{E}{P} \left[(1-\sigma) \frac{p(\omega)^{-\sigma}}{P} + \frac{\sigma w p(\omega)^{-1-\sigma}}{\phi P} \right]$$

We can immediately move one term to the other side of the equation and get rid of $\frac{E}{P}$ and $\frac{1}{P}$, which appear in both terms. That gives

$$(\sigma - 1) p(\omega)^{-\sigma} = \sigma \frac{w}{\phi} p(\omega)^{-1-\sigma}$$

We can multiply both sides by $p(\omega)^{-1-\sigma}$ and get $p(\omega)$ on one side, so we do that to get the much better-looking

$$(\sigma - 1) p(\omega) = \frac{\sigma w}{\phi}$$

which becomes

$$p(\omega) = \frac{\sigma}{\sigma - 1} \frac{w}{\phi}$$

Now we have an equation for the price. Returning to the original firm problem, we maximized profits as

$$\pi = \left(p(\omega) - \frac{w}{\phi} \right) q(\omega)$$

We first consider just revenue, dropping the $-\frac{w}{\phi}q(\omega)$ term. This becomes

$$r(\phi) \equiv p(\omega)q(\omega) = \frac{\sigma}{\sigma-1}\frac{w}{\phi}\left(\frac{p(\omega)}{P}\right)^{-\sigma}\frac{E}{P} = \left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)p(\omega)^{-\sigma}EP^{\sigma-1}$$

We then substitute for $p(\omega)$ again:

$$r(\phi) = \left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{-\sigma}EP^{\sigma-1} = EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma}$$

This is exactly the equation we want for revenue! Now we solve for profits:

$$\pi(\phi) = r(\phi) - \frac{w}{\phi}q(\omega) = EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma} - \frac{w}{\phi}p(\omega)^{-\sigma}EP^{\sigma-1}$$

and again re-substitute for $p(\omega)$ to get

$$\pi(\phi) = EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma} - \frac{w}{\phi}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{-\sigma}EP^{\sigma-1}$$

We multiply the second term by $1 = \left(\frac{\sigma}{\sigma-1}\right)\left(\frac{\sigma-1}{\sigma}\right)$, which allows us to rewrite it as

$$\frac{\sigma-1}{\sigma}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma}EP^{\sigma-1}$$

Thus we can write profits as

$$\pi(\phi) = \left[EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma}\right]\left(\frac{\sigma}{\sigma} - \frac{\sigma-1}{\sigma}\right) = \frac{1}{\sigma}\left[EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma}\right] = \frac{r(\phi)}{\sigma}$$

as required.

1.4

1.4.1 $\sigma \to \infty$

As the elasticity of substitution σ approaches ∞, goods should become infinitely substitutable — meaning that we move towards a case where we have a mass of firms measure one supplying identical goods. This is just perfect competition! While there is not an easy way to demonstrate the model limits to infinite competition here, we can verify this by looking at the quantity function:

$$\lim_{\sigma \to \infty} q(\omega) = \lim_{\sigma \to \infty} \left(\frac{p(\omega)}{P} \right)^{-\sigma} \frac{E}{P} = 0$$

We can interpret this result in two ways. First, if we assume all firms have the same productivity, each firm should produce an infintesimally small amount of the good in the limit — which limits to zero. Alternatively, if we think about firms drawing their productivities from some underlying distribution, then the most productive firm should be able to undercut the others on price and get all the business — which means that firm would have mass one and be the only firm producing, while the less productive firms would be driven out of the market in equilibrium and then produce a zero quantity of goods.

1.4.2 $\sigma \to 0$

Now we consider the case where σ approaches zero. The easiest way to think about this case is considering the relationship between two different goods:

$$\frac{q(\omega)}{q(\omega')} = \left(\frac{p(\omega)}{p(\omega')} \right)^{-1} = \left(\frac{p(\omega')}{p(\omega)} \right)$$

What this relationship tells us is that the consumer buys positive quantities of each good in relationship with their prices — effectively a case similar to the Cobb-Douglas production function. Effectively, this means a constant share of expenditure is assigned to each good, and the firm derives its desired production based on the relationship between the expenditure and its price:

$$q(\omega) = \left(\frac{p(\omega)}{P} \right)^{-1} \frac{E}{P} = \frac{E}{p(\omega)}$$

Of course, each firm is able to monopolistically set its own price and quantity, so this function becomes ill-behaved since the firm wants to produce as little as possible to minimize costs. Effectively, the quantity produced will limit to 0 as the firm makes money by raising prices arbitrarily high and producing an infinitesimal amount of the good, but the revenue will always be equivalent to the expenditure share allocated to the firm's good.

1.4.3 $\sigma \to 1$

Now we consider the case of $\sigma \to 0$. In this case, we have the relationship between two goods given by

$$\frac{q(\omega)}{q(\omega')} = \left(\frac{p(\omega)}{p(\omega')}\right)^0 = 1 \implies q(\omega) = q(\omega')$$

This relationship means that the consumer buys the same amount of each good — the case of perfect complements! In this case we should expect each firm to capture a quantity of goods equal to the lowest produced by any firm. This more or less shows up in the expression for quantities:

$$q(\omega) = \left(\frac{p(\omega)}{P}\right)^0 \frac{E}{P} = \frac{E}{P}$$

In the Leontief case, quantity is only determined by the consumer's total expenditures and the overall price level; the firm-specific productivity does not enter the firm's decision at all. We can also find that revenue is zero, but this is because with measure one of firms, each firm is producing the same infinitesimal quantity of their good, which means their revenue limits to zero.

2 Chaney (2008)

2.1

Under a Pareto distribution, the cumulative distribution function is given by

$$G(\phi) = 1 - \left(\frac{\phi_{min}}{\phi}\right)^k, \phi \geqslant \phi_{min}$$

We can find the density function by taking the derivative. This gives us

$$g(\phi) = \frac{dG(\phi)}{d\phi} = 0 - \frac{d}{d\phi}\left(\phi_{min}^{k}\phi^{-k}\right) = -\phi_{min}^{k}(-k)\phi^{-(k+1)}$$

Simplifying this expression, we derive the density function as

$$g(\phi) = \frac{k\phi_{min}^{k}}{\phi^{k+1}}$$

Now we want to find the probability that a productivity draw exceeds some ϕ_c. This is easiest to do using the techniques of survival analysis.[1] Assuming we have $\phi_c > \phi_{min}$ (so the answer isn't just 1), this is given by plugging ϕ_c into the survival function:

$$\mathbb{P}(\phi > \phi_c) = S(\phi_c) = 1 - G(\phi_c) = \left(\frac{\phi_{min}}{\phi_c}\right)^{k}$$

2.2

We want the probability distribution conditional on surviving the cutoff ϕ_d^*. To make the math slightly easier, let's write down the conditional survival function, then use it to derive the conditional probability distribution. The conditional probability survival function leads us directly to the conditional cumulative density function:

$$S(\phi|\phi > \phi_d^*) = \frac{S(\phi)}{S(\phi_d^*)} = \left(\frac{\frac{\phi_{min}}{\phi}}{\frac{\phi_{min}}{\phi_d^*}}\right)^{k} = \left(\frac{\phi_d^*}{\phi}\right)^{k} \implies G(\phi|\phi > \phi_d^*) = 1 - \left(\frac{\phi_c}{\phi}\right)^{k}$$

This replicates a well-known fact about Pareto random variables: the conditional probability distribution of a Pareto random variable is just a Pareto random variable with a different minimum value! Using this knowledge, we can follow our above derivation of the unconditional density to derive the conditional density by inspection:

$$g(\phi|\phi > \phi_d^*) = \frac{k\phi_d^*}{\phi^{k+1}}$$

[1] Survival functions, and particularly survival analysis involving Pareto distributions, appear all the time in actuarial work! Pareto random variables make for very useful loss models.

2.3

The conditional expectation is just the expectation of the conditional probability distribution, which is just our new Pareto distribution. The formula for the expectation is a property well-known enough that we could simply state it, but let's derive it by integrating the survival function.

$$\int_{\phi_d^*}^{\infty} S(\phi|\phi > \phi_d^*)d\phi = \int_{\phi_d^*}^{\infty} (\phi_d^*)^k (\phi)^{-k} \, d\phi = \left[(\phi_d^*)^k \left(\frac{1}{-k+1} \right) \phi^{-k+1} \right]_{\phi_d^*}^{\infty}$$

We assume that $k \geqslant 1$, which should be safe given the typical values chosen for k. This prevents the expectation from limiting to ∞. Then we can write

$$\mathbb{E}\left[\phi|\phi > \phi_d^*\right] = \left[(\phi_d^*)^k \left(\frac{1}{1-k} \right) \phi^{1-k} \right]_{\phi_d^*}^{\infty} = 0 - \frac{\phi_d^*}{1-k} = \frac{\phi_d^*}{k-1}$$

2.4

We now assume that the consumer has a CES demand function. From the results in the previous part, that immediately lets us write the general revenue function as

$$r(\phi) = EP^{\sigma-1} \left(\frac{\sigma}{\sigma-1} \frac{w}{\phi} \right)^{1-\sigma}$$

with profit given by $\pi(\phi) = \frac{r(\phi)}{\sigma} - cf$. Now we want to find revenue conditional on exceeding a cutoff ϕ_c. Let's rewrite the revenue function to find this:

$$\mathbb{E}\left[r(\phi) \mid \phi > \phi_c\right] = \mathbb{E}\left[EP^{\sigma-1} \left(\frac{\sigma}{\sigma-1} \frac{w}{\phi} \right)^{1-\sigma} \,\middle|\, \phi > \phi_c \right]$$

The only thing inside the expectation that is dependent on ϕ is ϕ itself, and it is separable from the rest of the function. That means we can write

$$\mathbb{E}\left[r(\phi) \mid \phi > \phi_c\right] = EP^{\sigma-1} \left(\frac{\sigma}{\sigma-1} w \right)^{1-\sigma} \mathbb{E}\left[\phi^{\sigma-1} \,\middle|\, \phi > \phi_c \right]$$

We already know that $\phi^{\sigma-1}$ is just a function taken over the new Pareto distribution with

minimum ϕ_c, so it remains to find this function. This can be found through survival analysis[2] as

$$\mathbb{E}\left[\phi^{\sigma-1}\right] = \int_{\phi_c}^{\infty} \left(\frac{d}{dx}\phi^{\sigma-1}\right) S(\phi)d\phi = \int_{\phi_c}^{\infty} (\sigma-1)\phi^{\sigma-2}\phi_c^k\phi^{-k}d\phi$$

We rewrite this integral as

$$\mathbb{E}\left[\phi^{\sigma-1}\right] = (\sigma-1)\phi_c^k \int_{\phi_c}^{\infty} \phi^{\sigma-2-k}d\phi = (\sigma-1)\phi_c^k \left[\frac{1}{\sigma-1-k}\phi^{\sigma-1-k}\right]_{\phi_c}^{\infty}$$

Assuming we have $k > \sigma - 1$ as given to us in the problem, the bracketed expression limits to zero as ϕ approaches ∞. Then we derive

$$\mathbb{E}\left[\phi^{\sigma-1}\right] = \frac{\sigma-1}{k-(\sigma-1)}\phi_c^{\sigma-1}$$

Note that this expectation should explode as $k \to \sigma - 1$; it is not well-defined for smaller values of k.

2.5

We now want to find the zero-profit cutoff rule for both domestic firms and exporters. The profit function is given for domestic firms by

$$\pi(\phi) = \frac{1}{\sigma}\left[EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}w\right)^{1-\sigma}\right]\mathbb{E}\left[\phi^{\sigma-1}\middle|\phi > \phi_d^*\right] - f_d$$

and for exporters as

$$\pi(\phi) = \frac{\tau}{\sigma}\left[E^*(P^*)^{\sigma-1}\left(\frac{\sigma}{\sigma-1}w\right)^{1-\sigma}\right]\mathbb{E}\left[\phi^{\sigma-1}\middle|\phi > \phi_x^*\right] - f_x$$

These expressions are easy enough to manipulate for ϕ_d^* and ϕ_x^* now that we have a formula for the conditional expectation term. Plugging it in, setting the profit functions

[2]Strangely enough, this result is *not* commonly taught in actuarial material? It seems like it would be extremely useful. The proof is straightforward; I found it on slide 10 here.

equal to zero, and isolating ϕ_d^* and ϕ_x^*, we find

$$\phi_d^* = \left(\frac{\sigma}{\sigma-1}\right)^{\frac{\sigma}{\sigma-1}} (w f_d (k-(\sigma-1)))^{\frac{1}{\sigma-1}} \left(\frac{E^{\frac{1}{\sigma-1}}}{P}\right)$$
$$\phi_x^* = \left(\frac{\sigma}{\sigma-1}\right)^{\frac{\sigma}{\sigma-1}} (\tau w f_x (k-(\sigma-1)))^{\frac{1}{\sigma-1}} \left(\frac{(E^*)^{\frac{1}{\sigma-1}}}{P^*}\right)$$

2.6

The fraction of domestic firms that are exporters is given by solving the survival function of the distribution conditional on producing domestically above the export cutoff:

$$S(\phi_x^*) = \left(\frac{\phi_d^*}{\phi_x^*}\right)^k$$

We can simplify this expression further before plugging in the expression we found above. Observe that we can rewrite our general profit functions as

$$\pi(\phi_d^*) = E P^{\sigma-1} C (\phi_d^*)^{\sigma-1} - f_d = 0 \qquad \implies C = \frac{f_d}{(\phi_d^*)^{\sigma-1} E P^{\sigma-1}}$$
$$\pi(\phi_x^*) = E^* (P^*)^{\sigma-1} \tau C (\phi_x^*)^{\sigma-1} - f_x = 0 \quad \implies f_x = \tau f_d \left(\frac{E^*}{E}\right) \left(\frac{P^*}{P}\right)^{\sigma-1} \left(\frac{\phi_x^*}{\phi_d^*}\right)^{\sigma-1}$$

This lets us write the export cutoff as a function of the domestic cutoff:

$$\phi_d^* = \left(\frac{E^*}{E}\right)^{\frac{1}{\sigma-1}} \left(\frac{P^*}{P}\right) \left(\tau \frac{f_d}{f_x}\right)^{\frac{1}{\sigma-1}} \phi_x^*$$

Then the survival function becomes

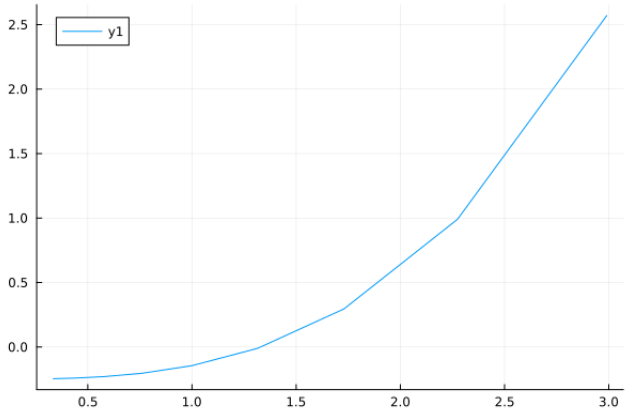
$$S(\phi_x^*) = \left(\frac{\phi_d^*}{\phi_x^*}\right)^k = \left[\left(\frac{E^*}{E}\right) \left(\frac{P^*}{P}\right)^{\sigma-1} \left(\tau \frac{f_d}{f_x}\right)\right]^{\frac{k}{\sigma-1}}$$

which will give us the fraction of domestically active firms that are exporters in terms of model primitives and market-specific values taken as given by the firm.

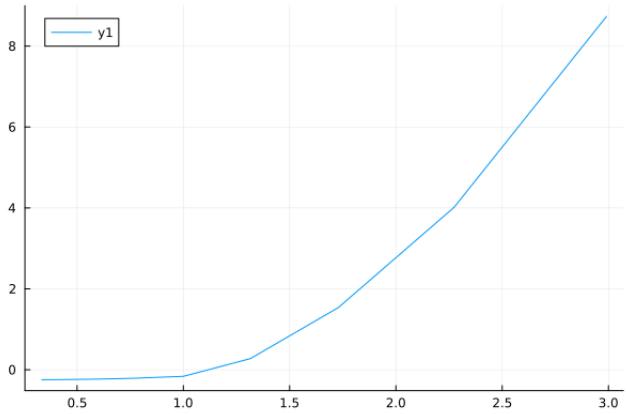
We see from this equation that more firms will become exporters if the foreign market demand $E^* (P^*)^{\sigma-1}$ becomes more valuable. Conversely, since τ is bounded between 0 and 1 — and raised to an exponent greater than 1 since $k > \sigma - 1$! — increasing τ will lower the percentage of firms that export. This all makes intuitive sense: if the foreign market becomes more profitable firms should be more willing to export, and if trade becomes harder firms should be less willing to export.

3 Hopenhayn on the Computer

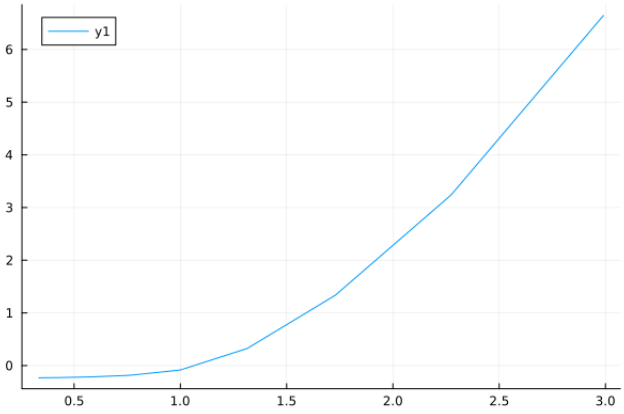
3.1



3.2



3.3



3.4

3.5

3.6

3.7

3.8